# A CONE-EVENT OBSTRUCTION TO ALMOST-SURE CONVERGENCE IN ERDŐS PROBLEM #521

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Abstract. Let

$$f_n(x) = \sum_{k=0}^{n} \varepsilon_k x^k, \qquad x \in \mathbb{R}, \qquad \varepsilon_k \in \{-1, 1\},$$

be the Rademacher Kac polynomial. Let $R_n$ be the number of real zeros of $f_n$, let $I_n$ be the number of zeros in $[-1, 1]$, and let $O_n$ be the number of zeros outside $[-1, 1]$. We strengthen the elementary cone-event obstruction from positive probability to an almost-sure statement:

$$\mathbb{P}\big(O_n = 0 \text{ for infinitely many } n\big) = 1.$$

Combining this with Do's strong law $I_n/\log n \to 1/\pi$ almost surely gives

$$\liminf_{n\to\infty} \frac{R_n}{\log n} = \frac{1}{\pi} \qquad \text{almost surely.}$$

Finally, reversal symmetry and a deterministic-subsequence argument give

$$\limsup_{n\to\infty} \frac{R_n}{\log n} \geq \frac{2}{\pi} \qquad \text{almost surely.}$$

Consequently $R_n/\log n$ does not converge almost surely. In particular, the almost-sure assertion $R_n/\log n \to 2/\pi$ in Erdős Problem #521 is false for the Rademacher model.

## 1. Setup and external input

All logarithms are natural. Throughout the paper, zeros are counted without multiplicity. Thus, if $P$ is a real polynomial and $A \subset \mathbb{R}$, then

$$N_P(A) := \#\{x \in A : P(x) = 0\}.$$

We work on a probability space carrying independent Rademacher random variables $(\varepsilon_k)_{k\geq 0}$, that is,

$$\mathbb{P}(\varepsilon_k = 1) = \mathbb{P}(\varepsilon_k = -1) = \frac{1}{2}, \qquad k \geq 0.$$

For each integer $n \geq 0$, define the random polynomial

$$f_n(x) = \sum_{k=0}^{n} \varepsilon_k x^k, \qquad x \in \mathbb{R}. \tag{1.1}$$

Set

$$R_n = N_{f_n}(\mathbb{R}), \qquad I_n = N_{f_n}([-1, 1]), \qquad O_n = N_{f_n}\big((-\infty, -1) \cup (1, \infty)\big). \tag{1.2}$$

Then $R_n = I_n + O_n$ for every $n$. Erdős Problem #521 asks whether, for Rademacher coefficients,

$$\frac{R_n}{\log n} \longrightarrow \frac{2}{\pi} \qquad \text{almost surely} \tag{1.3}$$

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as $n \to \infty$; see [4]. The only non-elementary input used below is the following theorem of Do [1].

**Theorem 1.1** (Do)**.** *Let $(\xi_k)_{k\geq 0}$ be independent and identically distributed (i.i.d.) real random variables with mean zero, variance one, and a bounded $(2+\eta)$-moment for some $\eta > 0$. Let $p_n(x) = \sum_{k=0}^{n} \xi_k x^k$. Then*

$$\frac{N_{p_n}([-1,1])}{\log n} \longrightarrow \frac{1}{\pi} \qquad \textit{almost surely.}$$

*In particular, this conclusion holds for the Rademacher coefficients in (1.1).*

## 2. Reversal and a deterministic positivity criterion

For $n \geq 0$, define the reversed polynomial

$$g_n(y) = y^n f_n(1/y) = \sum_{r=0}^{n} \varepsilon_{n-r} y^r, \qquad y \in \mathbb{R}. \tag{2.1}$$

For $y \neq 0$, the change of variables $x = 1/y$ gives a bijection between zeros of $g_n$ in $(-1,1)\setminus\{0\}$ and zeros of $f_n$ in $(-\infty,-1)\cup(1,\infty)$. Since $g_n(0) = \varepsilon_n \neq 0$, this is equivalently a bijection between zeros of $g_n$ in $(-1,1)$ and zeros of $f_n$ outside $[-1,1]$. Consequently

$$O_n = 0 \quad \Longleftrightarrow \quad g_n \text{ has no zero in } (-1,1). \tag{2.2}$$

**Lemma 2.1** (Two partial-sum positivity criterion)**.** *Let $n \geq 0$, and let*

$$P(x) = \sum_{r=0}^{n} a_r x^r, \qquad a_r \in \{-1,1\}.$$

*For $0 \leq k \leq n$, define*

$$S_k = \sum_{r=0}^{k} a_r, \qquad T_k = \sum_{r=0}^{k} (-1)^r a_r.$$

*If*

$$S_k \geq 0 \quad \textit{and} \quad T_k \geq 0 \qquad \textit{for every } 0 \leq k \leq n, \tag{2.3}$$

*then $P(x) > 0$ for every $x \in (-1,1)$.*

*Proof.* Taking $k = 0$ in (2.3) gives $a_0 = S_0 = T_0 \geq 0$. Since $a_0 \in \{-1,1\}$, we have $a_0 = 1$.

First let $0 < x < 1$. Abel summation gives, with the convention that an empty sum is zero when $n = 0$,

$$P(x) = (1-x)\sum_{k=0}^{n-1} S_k x^k + S_n x^n. \tag{2.4}$$

Every term on the right-hand side is nonnegative. If $n = 0$, then $P(x) = a_0 = 1$. If $n \geq 1$, the $k = 0$ term in (2.4) is $(1-x)S_0 = 1 - x > 0$. Thus $P(x) > 0$ on $(0,1)$.

Next let $-1 < x < 0$, and write $x = -y$ with $0 < y < 1$. Then

$$P(-y) = \sum_{r=0}^{n} (-1)^r a_r y^r.$$

Applying (2.4) to the coefficient sequence $((-1)^r a_r)_{0\leq r\leq n}$ gives

$$P(-y) = (1-y)\sum_{k=0}^{n-1} T_k y^k + T_n y^n > 0,$$

by the same argument. Finally, $P(0) = a_0 = 1$. Hence $P$ is strictly positive throughout $(-1, 1)$. □

For $n \geq 0$, define the cone event

$$\mathcal{C}_n = \left\{ \begin{array}{ll} \sum_{r=0}^{k} \varepsilon_{n-r} \geq 0 & \text{for every } 0 \leq k \leq n, \\ \sum_{r=0}^{k} (-1)^r \varepsilon_{n-r} \geq 0 & \text{for every } 0 \leq k \leq n. \end{array} \right\} \tag{2.5}$$

Lemma 2.1, applied to $P = g_n$, implies

$$\mathcal{C}_n \subseteq \{O_n = 0\}. \tag{2.6}$$

## 3. Cone probabilities with terminal margin

The next estimates quantify the probability of the two simultaneous partial-sum conditions in (2.5). The key point is not merely that this probability is of order $L^{-1}$ for a block of length $L$, but that the same lower bound survives after imposing a fixed terminal margin $B$, provided $L$ is at least a constant multiple of $(B+1)^2$. This margin will later absorb an arbitrary finite initial segment of coefficients.

For an integer $L \geq 1$ and a real number $B \geq 0$, let $q_L(B)$ be the probability that iid Rademacher signs $a_0, \ldots, a_{L-1}$ satisfy

$$\sum_{r=0}^{k} a_r \geq 0, \qquad \sum_{r=0}^{k} (-1)^r a_r \geq 0 \qquad (0 \leq k < L), \tag{3.1}$$

and also the terminal margin conditions

$$\sum_{r=0}^{L-1} a_r \geq B, \qquad \sum_{r=0}^{L-1} (-1)^r a_r \geq B. \tag{3.2}$$

Write $q_L = q_L(0)$. Since the reversed signs $(\varepsilon_n, \varepsilon_{n-1}, \ldots, \varepsilon_0)$ have the same law as $n+1$ i.i.d. Rademacher signs,

$$\mathbb{P}(\mathcal{C}_n) = q_{n+1}. \tag{3.3}$$

**Lemma 3.1** (One-dimensional survival with terminal height)**.** *Let $(\xi_i)_{i \geq 1}$ be i.i.d. Rademacher signs, and let*

$$S_0 = 0, \qquad S_j = \sum_{i=1}^{j} \xi_i \quad (j \geq 1).$$

*For $r \geq 0$ and $B \geq 0$, define*

$$u_r(B) = \mathbb{P}\big(S_j \geq 0 \text{ for all } 0 \leq j \leq r,\ S_r \geq B\big).$$

*There exist positive absolute constants $c_1, C_1, K_1$ such that*

$$u_r(0) \leq C_1 (r+1)^{-1/2} \qquad (r \geq 0), \tag{3.4}$$

*and*

$$u_r(B) \geq c_1 (r+1)^{-1/2} \tag{3.5}$$

*whenever $r \geq K_1 (B+1)^2$.*

*Proof.* We recall the elementary reflection estimate in a form that fixes the random walk convention. For an integer $\ell \geq 0$ with the same parity as $r$, the number of paths of length $r$ from 0 to $\ell$ that ever hit $-1$ is, by reflection at the first visit to $-1$, equal to the number of unrestricted paths of length $r$ from 0 to $-\ell-2$. Hence

$$\begin{aligned}\mathbb{P}\big(S_j \geq 0 \text{ for all } 0 \leq j \leq r,\ S_r = \ell\big) \\ = 2^{-r}\left[\binom{r}{(r+\ell)/2} - \binom{r}{(r+\ell+2)/2}\right],\end{aligned} \tag{3.6}$$

where binomial coefficients with nonintegral lower argument are interpreted as zero.

We now simplify the difference of binomial coefficients. For admissible $\ell$,

$$\binom{r}{(r+\ell)/2} - \binom{r}{(r+\ell+2)/2} = \binom{r}{k} - \binom{r}{k+1}, \quad \text{where } k = \frac{r+\ell}{2}.$$

Using the identity

$$\binom{r}{k} - \binom{r}{k+1} = \binom{r}{k}\left(1 - \frac{r-k}{k+1}\right) = \binom{r}{k}\frac{2k+1-r}{k+1},$$

and substituting $k = (r+\ell)/2$, we obtain

$$2k+1-r = \ell+1, \qquad k+1 = \frac{r+\ell+2}{2}.$$

Thus

$$\binom{r}{k} - \binom{r}{k+1} = \binom{r}{k}\cdot\frac{\ell+1}{(r+\ell+2)/2} = 2\frac{\ell+1}{r+1}\binom{r+1}{k+1}.$$

Multiplying by $2^{-r}$ gives

$$\mathbb{P}\big(S_j \geq 0 \text{ for all } 0 \leq j \leq r,\ S_r = \ell\big) = 2\frac{\ell+1}{r+1}\mathbb{P}(S_{r+1} = \ell+1). \tag{3.7}$$

Summing (3.6) over all admissible $\ell \geq 0$ gives

$$\mathbb{P}(S_j \geq 0 \text{ for all } 0 \leq j \leq r) = 2^{-r}\binom{r}{\lfloor r/2 \rfloor}.$$

The standard central-binomial bound $u_r(0) = 2^{-r}\binom{r}{\lfloor r/2\rfloor} \leq C_1(r+1)^{-1/2}$ proves (3.4).

We now give a quantitative lower bound (3.5) for the point probabilities of $S_{r+1}$ in the central region. For integers $m$ with the same parity as $r+1$, we have

$$\mathbb{P}(S_{r+1} = m) = 2^{-(r+1)}\binom{r+1}{(r+1+m)/2}.$$

Set $k = \frac{r+1+m}{2}$, so that $|m| \leq 3\sqrt{r}$ implies

$$\left|k - \frac{r+1}{2}\right| \leq \frac{3}{2}\sqrt{r}.$$

In particular, $k$ lies in the central window of the binomial distribution. By Stirling's formula, there exist absolute constants $c_* > 0$ and $r_0 \geq 1$ such that for all $r \geq r_0$ and all integers $k$ satisfying

$$\left|k - \frac{r+1}{2}\right| \leq 2\sqrt{r+1},$$

one has

$$\binom{r+1}{k} \geq c_*\frac{2^{r+1}}{\sqrt{r+1}}.$$

Therefore, for all $r \geq r_0$ and all integers $m$ with the same parity as $r+1$ satisfying $|m| \leq 3\sqrt{r}$, we obtain

$$\mathbb{P}(S_{r+1} = m) = 2^{-(r+1)}\binom{r+1}{k} \geq c_*(r+1)^{-1/2}. \tag{3.8}$$

Choose $K_1$ large enough that $r \geq K_1(B+1)^2$ implies both $r \geq r_0$ and $\sqrt{r} \geq B+1$. Sum (3.7) over admissible terminal heights $\ell \in [\sqrt{r}, 2\sqrt{r}]$. The interval $[\sqrt{r}, 2\sqrt{r}]$ has length $\sqrt{r}$, and exactly half of the integers in this interval have the same parity as $r$. Hence it contains at least $(1/2)\sqrt{r} - 1 \geq c\sqrt{r}$ admissible values of $\ell$ for some absolute constant $c > 0$ and all sufficiently large $r$. For each such $\ell$, we have $\ell \geq B$ and, by (3.8),

$$\mathbb{P}(S_{r+1} = \ell + 1) \geq c_*(r+1)^{-1/2}.$$

Moreover $(\ell+1)/(r+1) \geq c'(r+1)^{-1/2}$ on this interval. Therefore each admissible $\ell$ contributes at least $c''(r+1)^{-1}$ to (3.7), and there are at least $c\sqrt{r}$ such $\ell$. Thus

$$u_r(B) \geq \sum_{\substack{\ell \in [\sqrt{r}, 2\sqrt{r}] \\ \ell \equiv r \pmod 2}} 2\frac{\ell+1}{r+1}\mathbb{P}(S_{r+1} = \ell+1) \geq c_1(r+1)^{-1/2},$$

after decreasing $c_1$ if necessary. This proves the lemma. □

**Lemma 3.2** (Quadrant survival with terminal margin)**.** *Let $(W_j)_{j\geq 0}$ be the axis-step random walk on $\mathbb{Z}^2$ started at $W_0 = (0,0)$, with increments*

$$(1,0), \quad (-1,0), \quad (0,1), \quad (0,-1)$$

*each having probability $1/4$. For $N \geq 0$ and $B \geq 0$, define*

$$Q_N(B) = \mathbb{P}\big(W_j \in \mathbb{Z}^2_{\geq 0} \text{ for all } 0 \leq j \leq N,\ W_N \in [B,\infty)^2\big).$$

*There exist positive absolute constants $c_2, C_2, K_2$, depending only on the constants in Lemma 3.1, such that*

$$Q_N(0) \leq \frac{C_2}{N+1} \qquad (N \geq 0), \tag{3.9}$$

*and*

$$Q_N(B) \geq \frac{c_2}{N+1} \tag{3.10}$$

*whenever $N \geq K_2(B+1)^2$.*

*Proof.* Let $H_N$ be the number of horizontal moves among the first $N$ steps. Given $H_N = r$, the horizontal coordinate, observed only at horizontal moves, is a one-dimensional simple symmetric random walk of length $r$; similarly, the vertical coordinate is an independent one-dimensional simple symmetric random walk of length $N-r$. The two-dimensional walk stays in $\mathbb{Z}^2_{\geq 0}$ exactly when both of these one-dimensional walks stay nonnegative. Therefore

$$Q_N(B) = \sum_{r=0}^{N} \mathbb{P}(H_N = r)u_r(B)u_{N-r}(B), \tag{3.11}$$

where $u_r(B)$ is as in Lemma 3.1.

For the upper bound, Lemma 3.1 gives

$$Q_N(0) \leq C_1^2 \mathbb{E}\left[\frac{1}{\sqrt{(H_N+1)(N-H_N+1)}}\right]. \tag{3.12}$$

Let $A_N = \{N/4 \le H_N \le 3N/4\}$. On $A_N$, the integrand in (3.12) is at most $C/(N+1)$. On $A_N^c$, the integrand is at most 1, while the Chernoff bound for the binomial variable $H_N \sim \mathrm{Bin}(N, 1/2)$ gives $\mathbb{P}(A_N^c) \le Ce^{-cN}$. Since $e^{-cN} \le C'(N+1)^{-1}$ for all $N \ge 0$, (3.9) follows after increasing $C_2$.

For the lower bound, let $A'_N = \{N/3 \le H_N \le 2N/3\}$. There is an absolute constant $p_0 > 0$ such that $\mathbb{P}(A'_N) \ge p_0$ for all sufficiently large $N$.[1] In addition, by increasing $K_2$ we may assume this holds whenever $N \ge K_2(B+1)^2$. Choose $K_2 \ge 3K_1$. Then, on $A'_N$, both $H_N$ and $N - H_N$ are at least $K_1(B+1)^2$, so Lemma 3.1 gives

$$u_{H_N}(B)u_{N-H_N}(B) \ge c_1^2\big((H_N+1)(N-H_N+1)\big)^{-1/2} \ge \frac{c}{N+1}.$$

Restricting (3.11) to $A'_N$ yields

$$Q_N(B) \ge \frac{cp_0}{N+1},$$

which proves (3.10) after setting $c_2 = cp_0$. □

**Lemma 3.3** (Cone probability with terminal margin)**.** *There exist positive absolute constants $c_3, C_3, K_3$, depending only on the constants in Lemma 3.2, such that, for every $L \ge 1$,*

$$q_L \le \frac{C_3}{L}, \tag{3.13}$$

*and, for every $B \ge 0$,*

$$q_L(B) \ge \frac{c_3}{L} \tag{3.14}$$

*whenever $L \ge K_3(B+1)^2$.*

*Proof.* Group the signs into adjacent pairs. For each $j \ge 0$ for which the pair is available, define

$$\Delta_j = \left(\frac{a_{2j} + a_{2j+1}}{2}, \frac{a_{2j} - a_{2j+1}}{2}\right).$$

The four possible sign pairs $(a_{2j}, a_{2j+1})$ give the four axis increments $(1,0)$, $(-1,0)$, $(0,1)$, and $(0,-1)$, each with probability $1/4$. Let $N = \lfloor L/2 \rfloor$ and, for $0 \le j \le N$, set

$$X_j = \sum_{i=0}^{j-1} \Delta_i, \qquad X_0 = (0,0).$$

After $j$ completed pairs,

$$\sum_{r=0}^{2j-1} a_r = 2X_j^{(1)}, \qquad \sum_{r=0}^{2j-1} (-1)^r a_r = 2X_j^{(2)}. \tag{3.15}$$

If the cone conditions (3.1) hold, then $X_j \in \mathbb{Z}^2_{\ge 0}$ for all completed-pair times $0 \le j \le N$. Therefore, by Lemma 3.2,

$$q_L \le Q_N(0) \le \frac{C_2}{N+1} \le \frac{C_3}{L},$$

after increasing $C_3$ to cover all $L \ge 1$.

For the lower bound, let $b = \lceil B/2 \rceil$. We construct an explicit subevent of the event defining $q_L(B)$. First force

$$(a_0, a_1) = (1,1), \qquad (a_2, a_3) = (1,-1), \tag{3.16}$$

[1]By the central limit theorem, $H_N$ is concentrated around $N/2$ with standard deviation of order $\sqrt{N}$. Since the interval $[N/3, 2N/3]$ has width of order $N$, it contains a fixed positive fraction of the probability mass. Hence there exists an absolute constant $p_0 > 0$ such that $\mathbb{P}(A'_N) \ge p_0$ for all sufficiently large $N$.

which has probability 1/16. During these first four one-step times, both the ordinary and alternating partial sums are nonnegative, and after two completed pairs the skeleton is at $X_2 = (1, 1)$.

Assume now that $L$ is large enough that $N \geq 2$. Conditional on (3.16), require the remaining completed-pair skeleton $X_2, X_3, \ldots, X_N$ to stay in $(1, 1) + \mathbb{Z}^2_{\geq 0}$ and to satisfy

$$X_N \in [1 + b, \infty)^2.$$

The probability of this additional requirement is $Q_{N-2}(b)$.

If $L$ is even, the terminal sums are $2X_N^{(1)}$ and $2X_N^{(2)}$, both at least $2(1 + b) \geq B$. If $L$ is odd, there is one unpaired final sign; this sign can reduce either terminal sum by at most one, so the final terminal sums are still at least

$$2(1 + b) - 1 \geq B.$$

Thus the constructed event is contained in the event defining $q_L(B)$. Choose $K_3$ sufficiently large, in terms of $K_2$, so that $L \geq K_3(B + 1)^2$ implies $N - 2 \geq K_2(b + 1)^2$. Lemma 3.2 then gives

$$q_L(B) \geq \frac{1}{16} Q_{N-2}(b) \geq \frac{c}{N + 1} \geq \frac{c_3}{L},$$

after decreasing $c_3$ if necessary. This proves the lower bound. □

## 4. Upgrading positive probability to probability one

We now turn the block estimate from Lemma 3.3 into an almost-sure statement. The idea is to condition on an arbitrary initial segment $\varepsilon_0, \ldots, \varepsilon_M$ and then look for a much longer fresh reversed block $\varepsilon_n, \ldots, \varepsilon_{M+1}$. If the fresh block satisfies the two cone conditions with terminal margin $M + 1$, then the remaining fixed prefix can change either partial sum by at most $M + 1$ and therefore cannot make the full reversed sequence leave the cone.

For $M \geq 0$, let

$$\mathcal{F}_M = \sigma(\varepsilon_0, \ldots, \varepsilon_M), \qquad B_M = M + 1.$$

For $n > M$, put

$$L_{n,M} = n - M. \tag{4.1}$$

Thus the fresh block

$$\varepsilon_n, \varepsilon_{n-1}, \ldots, \varepsilon_{M+1}$$

has length $L_{n,M}$. There is no additional independent length parameter in the notation below: once $M$ and $n$ are fixed, $L_{n,M}$ is fixed by (4.1). In particular, when $M$ is fixed and $n$ varies, the block length varies with $n$.

Define $\mathcal{D}_n^{(M)}$ to be the event that this fresh reversed block satisfies the two cone inequalities and ends with both terminal sums at least $B_M$. Explicitly,

$$\mathcal{D}_n^{(M)} = \left\{ \begin{array}{ll} \sum_{r=0}^{k} \varepsilon_{n-r} \geq 0 & \text{for every } 0 \leq k < L_{n,M}, \\ \sum_{r=0}^{k} (-1)^r \varepsilon_{n-r} \geq 0 & \text{for every } 0 \leq k < L_{n,M}, \\ \sum_{r=0}^{L_{n,M}-1} \varepsilon_{n-r} \geq B_M, & \\ \sum_{r=0}^{L_{n,M}-1} (-1)^r \varepsilon_{n-r} \geq B_M. & \end{array} \right\} \tag{4.2}$$

**Lemma 4.1.** *For every $M \geq 0$ and every $n > M$,*

$$\mathcal{D}_n^{(M)} \subseteq \mathcal{C}_n.$$

*Proof.* Let $L = L_{n,M}$ and $B = B_M$. On $\mathcal{D}_n^{(M)}$, the first $L$ reversed signs, namely $\varepsilon_n, \ldots, \varepsilon_{M+1}$, already satisfy the two cone inequalities required in the definition of $\mathcal{C}_n$. Their terminal ordinary and alternating sums are both at least $B = M + 1$.

It remains to append the fixed prefix

$$\varepsilon_M, \varepsilon_{M-1}, \ldots, \varepsilon_0.$$

Any partial segment of this prefix has length at most $M + 1$, and hence its ordinary signed contribution has absolute value at most $M + 1 = B$. The same bound holds for the alternating contribution. Indeed, when the prefix is appended to the fresh block, the alternating signs start with parity $(-1)^L$, but multiplication by signs of modulus one does not change the bound on the absolute value of any partial sum. Therefore neither coordinate can be reduced by more than $B$ after the fresh block has built terminal margin $B$. All subsequent full reversed partial sums are nonnegative, and so $\mathcal{C}_n$ occurs. □

Since $\mathcal{D}_n^{(M)}$ depends only on the signs $\varepsilon_{M+1}, \ldots, \varepsilon_n$, it is independent of $\mathcal{F}_M$. By definition of $q_L(B)$,

$$\mathbb{P}(\mathcal{D}_n^{(M)} \mid \mathcal{F}_M) = q_{L_{n,M}}(B_M) \qquad (n > M). \tag{4.3}$$

The next lemma is the only dependence estimate needed for the second-moment argument. The notation is deliberately explicit: in the lemma below, $M$ is fixed, while $m$ and $n$ vary. We set $L_{m,M} = m - M$ and $L_{n,M} = n - M$.

**Lemma 4.2** (Conditional pair bound)**.** *For all integers $M < m < n$,*

$$\mathbb{P}(\mathcal{D}_m^{(M)} \cap \mathcal{D}_n^{(M)} \mid \mathcal{F}_M) \leq q_{L_{m,M}}(B_M) q_{n-m} \qquad \textit{almost surely.} \tag{4.4}$$

*Proof.* Let $E_{n,m}$ be the event that the block

$$\varepsilon_n, \varepsilon_{n-1}, \ldots, \varepsilon_{m+1}$$

of length $n - m$ satisfies the two cone inequalities with no terminal-margin requirement. Then $\mathbb{P}(E_{n,m}) = q_{n-m}$. Moreover, $E_{n,m}$ depends only on the signs $\varepsilon_{m+1}, \ldots, \varepsilon_n$, whereas $\mathcal{D}_m^{(M)}$ depends only on the signs $\varepsilon_{M+1}, \ldots, \varepsilon_m$. Hence these two events are independent conditionally on $\mathcal{F}_M$.

If both $\mathcal{D}_m^{(M)}$ and $\mathcal{D}_n^{(M)}$ occur, then the initial segment $\varepsilon_n, \dots, \varepsilon_{m+1}$ of the longer fresh block in $\mathcal{D}_n^{(M)}$ satisfies the two cone inequalities; that is, $E_{n,m}$ occurs. Therefore

$$\mathcal{D}_m^{(M)} \cap \mathcal{D}_n^{(M)} \subseteq \mathcal{D}_m^{(M)} \cap E_{n,m}.$$

Using conditional independence and (4.3), we get

$$\begin{aligned}\mathbb{P}(\mathcal{D}_m^{(M)} \cap \mathcal{D}_n^{(M)} \mid \mathcal{F}_M) &\le \mathbb{P}(\mathcal{D}_m^{(M)} \cap E_{n,m} \mid \mathcal{F}_M)\\ &= \mathbb{P}(\mathcal{D}_m^{(M)} \mid \mathcal{F}_M)\mathbb{P}(E_{n,m})\\ &= q_{L_{m,M}}(B_M) q_{n-m}.\end{aligned}$$

This proves (4.4). □

**Proposition 4.3.** *There exists a positive absolute constant $c_4$, depending only on the constants in Lemma 3.3, such that, for every $M \ge 0$,*

$$\mathbb{P}(\mathcal{C}_n \text{ occurs for infinitely many } n \mid \mathcal{F}_M) \ge c_4 \qquad \text{almost surely.} \tag{4.5}$$

*Proof.* Fix $M \ge 0$ and condition on $\mathcal{F}_M$. All probabilities and expectations below are therefore conditional on the already revealed initial segment encoded by $\mathcal{F}_M$. Let

$$B = B_M = M + 1.$$

Choose an integer $N$ so large that

$$N - M \ge K_3(B+1)^2 \qquad \text{and} \qquad N \ge 2M + 2, \tag{4.6}$$

where $K_3$ is the constant from Lemma 3.3. The first condition ensures that Lemma 3.3 may be applied to every index $n \in [N, N^2]$, since then

$$L_{n,M} = n - M \ge N - M \ge K_3(B+1)^2.$$

The second condition will be used later to compare $n - M$ and $n$ uniformly.

Define

$$Y_{N,M} = \sum_{n=N}^{N^2} \mathbf{1}_{\mathcal{D}_n^{(M)}}.$$

Thus $Y_{N,M}$ counts how many of the events $\mathcal{D}_n^{(M)}$ occur in the block of indices $N \le n \le N^2$.

We first estimate the first moment from below. By the definition of $Y_{N,M}$, by (4.3), and by Lemma 3.3,

$$\begin{aligned}\mathbb{E}(Y_{N,M} \mid \mathcal{F}_M) &= \sum_{n=N}^{N^2} \mathbb{P}(\mathcal{D}_n^{(M)} \mid \mathcal{F}_M)\\ &= \sum_{n=N}^{N^2} q_{L_{n,M}}(B)\\ &\ge c_3 \sum_{n=N}^{N^2} \frac{1}{L_{n,M}}\\ &= c_3 \sum_{n=N}^{N^2} \frac{1}{n-M}.\end{aligned} \tag{4.7}$$

Since $M \geq 0$, we have $n - M \leq n$, and hence

$$\frac{1}{n-M} \geq \frac{1}{n}.$$

Therefore

$$\sum_{n=N}^{N^2} \frac{1}{n-M} \geq \sum_{n=N}^{N^2} \frac{1}{n}.$$

The latter is a harmonic sum over the interval $[N, N^2]$. Comparing it with an integral gives

$$\sum_{n=N}^{N^2} \frac{1}{n} \geq \int_N^{N^2+1} \frac{dx}{x} = \log(N^2+1) - \log N \geq \log N.$$

Consequently, after decreasing the absolute constant if necessary,

$$\mathbb{E}(Y_{N,M} \mid \mathcal{F}_M) \geq c \log N.$$

This gives a lower bound of $\mathbb{E}(Y_{N,M} \mid \mathcal{F}_M)$.

Here and below, $c, C$ denote positive absolute constants depending only on $c_3, C_3, K_3$; they are not allowed to depend on $M$, on the realized initial segment, or on $N$.

We also need a corresponding upper bound on the first moment. Since $q_L(B) \leq q_L \leq C_3/L$, (4.3) gives

$$\begin{aligned} \mathbb{E}(Y_{N,M} \mid \mathcal{F}_M) &= \sum_{n=N}^{N^2} q_{L_{n,M}}(B) \\ &\leq C \sum_{n=N}^{N^2} \frac{1}{n-M}. \end{aligned} \tag{4.8}$$

Now the condition $N \geq 2M + 2$ implies $M \leq (N-2)/2$. Hence, for every $n \geq N$,

$$n - M \geq n - \frac{N}{2} \geq \frac{n}{2}.$$

Thus

$$\frac{1}{n-M} \leq \frac{2}{n}.$$

Therefore

$$\sum_{n=N}^{N^2} \frac{1}{n-M} \leq 2 \sum_{n=N}^{N^2} \frac{1}{n} \leq C \log N,$$

again by the standard harmonic-sum estimate. Hence

$$\mathbb{E}(Y_{N,M} \mid \mathcal{F}_M) \leq C \log N.$$

We next bound the second moment. Expanding the square gives

$$Y_{N,M}^2 = \sum_{n=N}^{N^2} \mathbf{1}_{\mathcal{D}_n^{(M)}} + 2 \sum_{N \leq m < n \leq N^2} \mathbf{1}_{\mathcal{D}_m^{(M)}} \mathbf{1}_{\mathcal{D}_n^{(M)}}.$$

Taking conditional expectations and using Lemma 4.2, we obtain

$$\begin{aligned}\mathbb{E}(Y_{N,M}^2 \mid \mathcal{F}_M) &\le \mathbb{E}(Y_{N,M} \mid \mathcal{F}_M) + 2 \sum_{N \le m < n \le N^2} q_{L_{m,M}}(B) q_{n-m} \\ &\le C \log N + C \sum_{N \le m < n \le N^2} \frac{1}{m-M} \frac{1}{n-m}. \end{aligned} \tag{4.9}$$

In the last line we used the upper bounds

$$q_{L_{m,M}}(B) \le \frac{C}{m-M}, \qquad q_{n-m} \le \frac{C}{n-m}.$$

We now estimate the double sum. For fixed $m$, write

$$h = n - m.$$

As $n$ ranges from $m+1$ to $N^2$, the variable $h$ ranges from 1 to $N^2 - m$. Hence

$$\mathbb{E}(Y_{N,M}^2 \mid \mathcal{F}_M) \le C \log N + C \sum_{m=N}^{N^2-1} \frac{1}{m-M} \sum_{h=1}^{N^2-m} \frac{1}{h}. \tag{4.10}$$

The inner sum is again harmonic. Since $N^2 - m \le N^2$, we have

$$\sum_{h=1}^{N^2-m} \frac{1}{h} \le \sum_{h=1}^{N^2} \frac{1}{h} \le C \log(N^2) \le C \log N.$$

Thus

$$\sum_{m=N}^{N^2-1} \frac{1}{m-M} \sum_{h=1}^{N^2-m} \frac{1}{h} \le C \log N \sum_{m=N}^{N^2-1} \frac{1}{m-M}.$$

Using again $N \ge 2M+2$, we have $m - M \ge m/2$ for all $m \ge N$, and therefore

$$\sum_{m=N}^{N^2-1} \frac{1}{m-M} \le 2 \sum_{m=N}^{N^2-1} \frac{1}{m} \le C \log N.$$

Combining these estimates gives

$$\sum_{m=N}^{N^2-1} \frac{1}{m-M} \sum_{h=1}^{N^2-m} \frac{1}{h} \le C (\log N)^2.$$

Substituting this into (4.10), and absorbing the lower order term $C \log N$ into $C(\log N)^2$, yields

$$\mathbb{E}(Y_{N,M}^2 \mid \mathcal{F}_M) \le C (\log N)^2. \tag{4.11}$$

We now apply the Paley–Zygmund inequality, conditionally on $\mathcal{F}_M$. Equivalently, by conditional Cauchy–Schwarz,

$$\mathbb{E}(Y_{N,M} \mid \mathcal{F}_M) = \mathbb{E}(Y_{N,M} \mathbf{1}_{\{Y_{N,M}>0\}} \mid \mathcal{F}_M) \le \mathbb{E}(Y_{N,M}^2 \mid \mathcal{F}_M)^{1/2} \mathbb{P}(Y_{N,M} > 0 \mid \mathcal{F}_M)^{1/2}.$$

Rearranging gives

$$\mathbb{P}(Y_{N,M} > 0 \mid \mathcal{F}_M) \ge \frac{\mathbb{E}(Y_{N,M} \mid \mathcal{F}_M)^2}{\mathbb{E}(Y_{N,M}^2 \mid \mathcal{F}_M)}.$$

Using the first-moment lower bound and the second-moment upper bound, we get

$$\mathbb{P}(Y_{N,M} > 0 \mid \mathcal{F}_M) \ge \frac{c^2 (\log N)^2}{C (\log N)^2} \ge c_4, \tag{4.12}$$

where $c_4 > 0$ is an absolute constant. Importantly, $c_4$ is independent of $M$, of the realized initial segment, and of $N$.

The estimate (4.12) holds for every sufficiently large $N$ satisfying (4.6). For $N \geq 1$, define

$$A_{N,M} = \{\mathcal{D}_n^{(M)} \text{ occurs for some } n \geq N\}.$$

The events $A_{N,M}$ are decreasing in $N$: if we require the first possible occurrence to happen after a later time, the event becomes smaller. Moreover, if $Y_{N,M} > 0$, then some $\mathcal{D}_n^{(M)}$ occurs with $N \leq n \leq N^2$, and therefore certainly some $\mathcal{D}_n^{(M)}$ occurs with $n \geq N$. Thus

$$\{Y_{N,M} > 0\} \subseteq A_{N,M}.$$

Consequently,

$$\mathbb{P}(A_{N,M} \mid \mathcal{F}_M) \geq c_4$$

for all sufficiently large $N$.

Since the events $A_{N,M}$ decrease as $N$ increases, conditional continuity from above gives

$$\mathbb{P}\left(\bigcap_{N=1}^{\infty} A_{N,M} \,\middle|\, \mathcal{F}_M\right) = \lim_{N\to\infty} \mathbb{P}(A_{N,M} \mid \mathcal{F}_M).$$

But

$$\bigcap_{N=1}^{\infty} A_{N,M} = \{\mathcal{D}_n^{(M)} \text{ occurs for infinitely many } n\}.$$

Indeed, belonging to every $A_{N,M}$ means that no matter how far out one looks, there is still some occurrence of $\mathcal{D}_n^{(M)}$ beyond that point, which is exactly infinite occurrence. Therefore

$$\mathbb{P}(\mathcal{D}_n^{(M)} \text{ occurs for infinitely many } n \mid \mathcal{F}_M) \geq c_4.$$

Finally, Lemma 4.1 implies that whenever $\mathcal{D}_n^{(M)}$ occurs infinitely often, the events $\mathcal{C}_n$ also occur infinitely often. Hence

$$\{\mathcal{D}_n^{(M)} \text{ occurs for infinitely many } n\} \subseteq \{\mathcal{C}_n \text{ occurs for infinitely many } n\}.$$

Thus

$$\mathbb{P}(\mathcal{C}_n \text{ occurs for infinitely many } n \mid \mathcal{F}_M) \geq c_4,$$

which proves (4.5). □

**Theorem 4.4.** *Almost surely,*

$$\mathcal{C}_n \textit{ occurs for infinitely many } n.$$

*Consequently,*

$$O_n = 0 \textit{ for infinitely many } n \qquad \textit{almost surely.}$$

*Proof.* Let

$$E = \{\mathcal{C}_n \text{ occurs for infinitely many } n\}.$$

By Proposition 4.3, for every fixed integer $M \geq 0$,

$$\mathbb{P}(E \mid \mathcal{F}_M) \geq c_4 \qquad \text{almost surely.}$$

For each fixed $M$, this inequality may fail only on a null event. Since there are only countably many values of $M$, we may take the union of these exceptional null events over all $M$. This union is still a null event. Therefore, outside a single null event, the lower bound

$$\mathbb{P}(E \mid \mathcal{F}_M) \geq c_4$$

holds simultaneously for every $M \geq 0$.

The sigma-fields $\mathcal{F}_M$ increase to

$$\mathcal{F}_\infty := \sigma(\varepsilon_0, \varepsilon_1, \ldots).$$

The event $E$ is determined by the whole sequence, and hence

$$E \in \mathcal{F}_\infty.$$

By Levy's zero-one law, for conditional expectations,

$$\mathbb{P}(E \mid \mathcal{F}_M) \longrightarrow \mathbf{1}_E \qquad \text{almost surely.}$$

Now work on the full-probability event on which both facts hold: the lower bound

$$\mathbb{P}(E \mid \mathcal{F}_M) \geq c_4 \quad \text{for all } M$$

and the convergence

$$\mathbb{P}(E \mid \mathcal{F}_M) \to \mathbf{1}_E.$$

Taking the limit in the lower bound gives

$$\mathbf{1}_E \geq c_4 \qquad \text{almost surely.}$$

Since $c_4 > 0$ and $\mathbf{1}_E$ can only take the values 0 and 1, this forces

$$\mathbf{1}_E = 1 \qquad \text{almost surely.}$$

Thus

$$\mathbb{P}(E) = 1,$$

which means that $\mathcal{C}_n$ occurs for infinitely many $n$ almost surely.

Finally, by the inclusion (2.6), each occurrence of $\mathcal{C}_n$ implies $O_n = 0$. Therefore, if $\mathcal{C}_n$ occurs for infinitely many $n$, then $O_n = 0$ also occurs for infinitely many $n$. Hence

$$O_n = 0 \text{ for infinitely many } n \qquad \text{almost surely.}$$

□

## 5. The liminf is $1/\pi$ almost surely

**Theorem 5.1.** *For Rademacher Kac polynomials,*

$$\liminf_{n\to\infty} \frac{R_n}{\log n} = \frac{1}{\pi} \qquad \textit{almost surely.}$$

*Proof.* Since $R_n = I_n + O_n$ and $O_n \geq 0$, Theorem 1.1 gives

$$\liminf_{n\to\infty} \frac{R_n}{\log n} \geq \lim_{n\to\infty} \frac{I_n}{\log n} = \frac{1}{\pi} \qquad \text{almost surely.}$$

Conversely, by Theorem 4.4, almost surely there exists a random subsequence $n_j \to \infty$ such that $O_{n_j} = 0$ for every $j$. Along this subsequence,

$$\frac{R_{n_j}}{\log n_j} = \frac{I_{n_j}}{\log n_j} \longrightarrow \frac{1}{\pi} \qquad \text{almost surely,}$$

again by Theorem 1.1. Thus the liminf is at most $1/\pi$ almost surely, and the theorem follows. □

## 6. A deterministic high subsequence and nonconvergence

The preceding section gives arbitrarily low values of $R_n/\log n$ along random subsequences. To prove nonconvergence, it remains to show that along some deterministic subsequence the outside zeros contribute the missing $1/\pi$. This uses only reversal symmetry and the strong law on $[-1,1]$.

**Lemma 6.1.** *The outside root count satisfies*

$$\frac{O_n}{\log n} \longrightarrow \frac{1}{\pi} \qquad \textit{in probability.}$$

*Consequently, there exists a deterministic subsequence $n_j \to \infty$ such that*

$$\frac{O_{n_j}}{\log n_j} \longrightarrow \frac{1}{\pi} \qquad \textit{almost surely.}$$

*Proof.* The coefficient vector $(\varepsilon_n, \varepsilon_{n-1}, \dots, \varepsilon_0)$ has the same law as $(\varepsilon_0, \varepsilon_1, \dots, \varepsilon_n)$. By the reversal relation (2.1), $O_n$ has the same distribution as $N_{f_n}((-1,1))$. Because zeros are counted without multiplicity,

$$|N_{f_n}((-1,1)) - N_{f_n}([-1,1])| \le 2.$$

Theorem 1.1 implies $I_n/\log n \to 1/\pi$ almost surely, hence in probability. The preceding deterministic bound by $2/\log n$ therefore gives

$$\frac{O_n}{\log n} \longrightarrow \frac{1}{\pi} \qquad \text{in probability.}$$

Choose $n_j$ recursively, with $n_j > n_{j-1}$, so that

$$\mathbb{P}\left(\left|\frac{O_{n_j}}{\log n_j} - \frac{1}{\pi}\right| > \frac{1}{j}\right) \le 2^{-j}.$$

This is possible because of the convergence in probability just proved. Borel–Cantelli then gives $O_{n_j}/\log n_j \to 1/\pi$ almost surely. □

**Theorem 6.2.** *For Rademacher Kac polynomials,*

$$\limsup_{n\to\infty} \frac{R_n}{\log n} \ge \frac{2}{\pi} \qquad \textit{almost surely.}$$

*In particular, $R_n/\log n$ does not converge almost surely. Hence the almost-sure assertion* (1.3) *is false.*

*Proof.* Let $(n_j)$ be the deterministic subsequence from Lemma 6.1. By Theorem 1.1,

$$\frac{I_{n_j}}{\log n_j} \to \frac{1}{\pi} \qquad \text{almost surely,}$$

and by Lemma 6.1,

$$\frac{O_{n_j}}{\log n_j} \to \frac{1}{\pi} \qquad \text{almost surely.}$$

Therefore

$$\frac{R_{n_j}}{\log n_j} = \frac{I_{n_j} + O_{n_j}}{\log n_j} \to \frac{2}{\pi} \qquad \text{almost surely.}$$

Thus the limsup is at least $2/\pi$ almost surely. Theorem 5.1 gives the almost-sure liminf $1/\pi$. Since $1/\pi < 2/\pi$, the sequence $R_n/\log n$ cannot converge almost surely; in particular, it cannot converge almost surely to $2/\pi$. □

*Remark* 6.3. The argument proves more than a positive-probability obstruction. It proves

$$O_n = 0 \text{ for infinitely many } n \text{ almost surely}, \qquad \liminf_{n\to\infty} R_n/\log n = 1/\pi \text{ almost surely},$$

and nonconvergence of $R_n/\log n$ almost surely. It does not identify the exact value of the limsup; Theorem 6.2 only gives the lower bound $\limsup R_n/\log n \geq 2/\pi$.

## Use of AI Tools

Portions of this work were developed with the assistance of ChatGPT-5.5-Pro. It was used to help explore ideas, generate preliminary drafts, and refine exposition. All mathematical arguments, results, and conclusions have been carefully reviewed and verified by the author, who takes full responsibility for the content of the paper. Any limitations or errors remain solely the responsibility of the author.

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